

AI61201: Visual Computing with AI/ML

Programming Assignment 3 — Frequency-Domain Image Processing

Autumn 2026 | Total: 20 Marks

This assignment is designed to assess practical understanding of 2D DFT/IDFT, frequency-domain filtering, Fourier-domain image translation/rotation, and separable DCT. You must show the requested visualizations, numerical checks, equations/transformations used, and concise interpretations—not only code.

Learning Objectives

- Interpret a centered log-magnitude DFT spectrum and identify periodic interference frequencies.
- Design and apply a Gaussian band-reject filter in the frequency domain.
- Use DFT magnitude/phase properties to translate and rotate an image.
- Implement the 2D DCT separably from first principles and verify it against OpenCV.
- Quantify radial DCT energy compaction and reconstruct an image using only low-radial-frequency coefficients.

General Instructions

- Use grayscale images for all experiments as the transformations mentioned are for single channel system.
- For Gaussian Band Reject Filtering, use the supplied corrupted image: `assets/interference_input.png`.
- For the DCT section, use the supplied image: `assets/landscape_input.png`.
- For the translation and rotation experiments (Q2 and Q3), use `assets/dct_input.png` (the same shapes image used to illustrate the DCT formula previously; independent from the Q4 image).
- You may use NumPy, OpenCV (cv2), and Matplotlib. Do not use a ready-made DCT implementation for the from-scratch DCT; `cv2.dct` is permitted only for verification.
- For the DFT sections, follow the function requirements stated in each question: OpenCV for the Gaussian-filtering DFT/IDFT and NumPy `fft2` for the translation/rotation section.
- All figures must have meaningful titles, axis labels where appropriate, and legends/colorbars when needed.
- Report important parameter choices (cut-offs, bandwidth, translation amounts, radial threshold, etc.) and justify them.
- Use the supplied images or equivalent TA-provided images only; do not silently replace the input with an unrelated image.

Input Data Provided

1. `interference_input.png` — grayscale image corrupted by a periodic sinusoidal interference pattern.

2. dct_input.png — suitable grayscale image for DCT and energy-compaction analysis.

Mark Distribution

Component	Marks
Q1. Gaussian Band Reject Filtering	6
Q1(a) DFT + centered log-magnitude spectrum	1.0
Q1(b) Identify interference frequency from spectrum	1.5
Q1(c) Design and display Gaussian band-reject filter	1.5
Q1(d) Frequency-domain filtering	1.0
Q1(e) IDFT and restored image	1.0
Q2. Translation using DFT	4
Q2(a) DFT magnitude + phase, zero-centered	1.0
Q2(b) Derive/apply translation transformation	1.0
Q2(c) IDFT and display translated image	1.0
Q2(d) Explain why magnitude/phase is or is not changed	1.0
Q3. Rotation by 180° using DFT	3
Q3(a) Transformation of DFT for 180° rotation	2.0
Q3(b) IDFT and display rotated image	1.0
Q4. Discrete Cosine Transform	7
Q4(a) Separable DCT from scratch	2.0
Q4(b) Verify with cv2.dct	1
Q4(c) Log visualization of DCT coefficients	0.5
Q4(d) Radial energy $E(r)$ + plot	0.5
Q4(e) Find/report threshold retaining 97.5% energy	1.0
Q4(f) Zero coefficients beyond threshold, inverse DCT, absolute difference + explanation	2.0
TOTAL	20

Question 1 — Gaussian Band Reject Filtering [6 Marks]

Input: I1.png

The input image for this question has the form $g(x, y) = f(x, y) + A \sin \left[2\pi \left(\frac{u_0 x}{M} + \frac{v_0 y}{N} \right) \right]$, where

$f(x, y)$ is the clean grayscale image and $g(x, y)$ is the corrupted image.

Use the supplied periodic-interference image and perform all operations using OpenCV DFT/IDFT functions.

1(a) [1.0] Compute the 2D DFT using OpenCV. Shift the spectrum so that the zero-frequency component is at the center and display $\log(1 + |F(u, v)|)$

1(b) [1.5] From the centered magnitude spectrum, identify the frequency coordinates of the periodic interference. Explain how the symmetric peaks correspond to a real sinusoidal interference. Report the estimated frequency coordinates (u_0, v_0) , using a clearly stated frequency-coordinate convention.

1(c) [1.5] Design a Gaussian band-reject filter with suitable low/high cut-off frequencies (or equivalently a center frequency and bandwidth). Display the filter as an image. Explain why your chosen pass/reject region removes the interference while preserving as much image content as possible.

1(d) [1.0] Apply the designed Gaussian band-reject filter in the frequency domain. Show the filtered spectrum and/or the filtered complex DFT before IDFT, and briefly explain the operation.

1(e) [1.0] Take the 2D IDFT using OpenCV functions and display the restored image. Compare it qualitatively with the corrupted image and comment on residual interference and image-detail loss.

Question 2 — Translating an Image by Manipulating Its DFT [4 Marks]

Input: I2.png

Use NumPy's `np.fft.fft2` implementation. Let the desired horizontal and vertical translations be $x_0 = 25$ (i.e. shifted right by 25 pixels) and $y_0 = -15$ (i.e. shifted upwards by 15 pixels)..

2(a) [1.0] Compute the 2D DFT and display the zero-centered log-magnitude spectrum and phase spectrum.

2(b) [1.0] Apply the appropriate Fourier-domain transformation to translate the image horizontally by x_0 and vertically by y_0 . Decide whether the magnitude, phase, or both must be modified. State the mathematical transformation and implement it.

2(c) [1.0] Take the IDFT of the transformed DFT and display the translated image. Verify visually that the requested displacement occurred.

2(d) [1.0] Explain, using the polar form $F(u, v) = |F(u, v)| e^{j\phi(u,v)}$, why the transformation you applied changes phase while leaving the magnitude unchanged (apart from numerical effects).

Question 3 — Rotating an Image by 180° using Its DFT [3 Marks]

Input: I2.png

3(a) [2.0] Apply a suitable transformation to the DFT to obtain a 180° spatial rotation. Decide whether the magnitude spectrum, phase spectrum, or both must be transformed. State the mathematical transformation you use and explain the discrete-coordinate convention carefully.

3(b) [1.0] Take the IDFT of the transformed DFT and display the 180°-rotated image. Confirm that the result corresponds to a spatial 180° rotation.

Question 4 — Discrete Cosine Transform [7 Marks]

Input: I3.png

Use the supplied DCT image. Implement the DCT from scratch in a separable manner.

For an $M \times N$ image, use the following definition:

$$C(u, v) = \alpha(u)\alpha(v) \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cos \left[\frac{\pi(2x+1)u}{2M} \right] \cos \left[\frac{\pi(2y+1)v}{2N} \right]$$

$$\alpha(u) = \begin{cases} \sqrt{\frac{1}{M}}, & u = 0 \\ \sqrt{\frac{2}{M}}, & u \neq 0 \end{cases}, \alpha(v) = \begin{cases} \sqrt{\frac{1}{N}}, & v = 0 \\ \sqrt{\frac{2}{N}}, & v \neq 0 \end{cases}$$

The inverse DCT is given by $f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} \alpha(u)\alpha(v) C(u, v) \cos \left[\frac{\pi(2x+1)u}{2M} \right] \cos \left[\frac{\pi(2y+1)v}{2N} \right]$

4(a) [2.0] Implement the 2D DCT from scratch in a separable manner. A recommended structure is: apply a 1D DCT to every row, then apply the same 1D DCT to every column. Do not call `cv2.dct` inside your from-scratch function.

4(b) [0.5] Verify your implementation against OpenCV's `cv2.dct`. Report the maximum absolute difference (or another clearly defined numerical error metric) and state whether the implementations agree within numerical precision.

4(c) [0.5] Visualize the DCT coefficient matrix after a logarithmic transformation, e.g. $\log(1+|C(u,v)|)$.

4(d) [1] For each radial frequency r , calculate the fraction of total DCT energy contained inside that radius:

$$E(r) = \frac{\sum_{u,v: u^2+v^2 \leq r^2} |C(u, v)|^2}{\sum_{u,v} |C(u, v)|^2}. \text{ Plot } E(r) \text{ as a function of } r.$$

4(e) [1.0] Select the smallest suitable radial threshold r such that at least 97.5% of the total DCT energy is retained. Report the threshold value and the corresponding retained energy.

4(f) [2.0] Set all DCT coefficients with r greater than the threshold to zero and perform the inverse DCT. Report the fraction of all DCT coefficients that are thus set to 0. Reconstruct the image by performing inverse DCT. Display the reconstructed image and the absolute difference between the original and the reconstructed images. Explain why the absolute difference is high in certain regions of the image, relating your explanation to edges, fine details, and discarded high-frequency coefficients.

Submission Guidelines

- A .py file and .ipynb file containing code, outputs, figures, numerical values, and short explanations for every subpart should be submitted as a zip named in the format :- <Roll_number>_Assignment_03
- All requested figures must be visible in the submission that is executed code snippets; do not submit code that requires the evaluator to rerun it just to see results.
- Clearly label each answer Q1(a), Q1(b), ..., Q4(f).
- State parameter choices and image dimensions wherever they affect your calculations.
- The TA will evaluate both implementation correctness and interpretation.
- The TA responsible for this assignment is Aishik Das (email: dasaishik25@kgpian.iitkgp.ac.in). Please contact the TA for any clarifications.